Distributions (Generalized Functions)

Asked 10 years, 6 months ago Modified 10 years, 6 months ago Viewed 455 times

2

Why is a distribution defined in terms of the inequality

$$|\langle \Gamma, \psi \rangle| \leq C \sum_{|\alpha| \leq N} \sup_{x \in S} |\partial^{\alpha} \psi|$$

for all $\psi \in C_c^{\infty}(\Omega)$, $S \subset \Omega$, S compact, K some constant, α a multi-index. And the existence of such constants C, N.

Now why is this definition written the way it is? Does it imply something about continuity of the distribution Γ with respect to ψ? If so, what?

Obviously something like

$$\Gamma(\psi) = \int_{\Omega} |f\psi| \ d\mu(x)$$

satisfies this if we let

$$K = \mu(S) \int_{\Omega} |f| \ d\mu(x)$$

so I see why this definition is satisfied. But why is it *necessary*?

real-analysis analysis distribution-theory

Share Cite

edited Oct 8, 2014 at 1:11

asked Oct 7, 2014 at 5:05



Anthony Peter
6,895 ● 2 ■ 41 ▲ 92

It is not necessary, but it has proven to be very useful. And your guess is correct, it is for the continuity of Γ with respect to ψ. For something like $\int \psi_n f \to \int \psi f$ you also need special convergence properties of $\psi_n \to \psi$, namely the uniform convergence. This uniform convergence is not necessary, but has proven to be very useful. BTW, you should have a look at your definition and the dependencies of your S and N ... – Vobo Oct 7, 2014 at 6:02

Add a comment

Answer

Sorted by: Highest score (default)

To answer this properly, one has to talk about the **reason** for introducing distributions first.

One of the main reasons for this is to be able to define a sort of generalized derivative for objects which are to "singular" to possess derivatives in a classical sense. For example, the functions

$$f(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0 \end{cases}$$

has no derivative (at 0) in the classical sense, but we will see below that it has one in the sense of distributions.

First, let us note (in dimension 1 for simplicity), that if $f \in C^1(\mathbb{R})$ and $g \in C_c^\infty(\mathbb{R})$ (we could take any interval), then partial integration yields

$$\begin{aligned} \int_{\mathbb{R}} f'(x)g(x)\,dx &= f(b)g(b) - f(a)g(a) - \int_a^b f(x)g'(x)dx \\ &= -\int_{\mathbb{R}} f(x)g'(x)\,dx, \end{aligned}$$

where $a < b$ are chosen such that $\operatorname{supp}(g) \subset (a, b)$.

The first idea is now to **identify** the function f with the functional

$$\varphi_f : C_c^\infty(\mathbb{R}) \to \mathbb{K}, g \mapsto \int_{\mathbb{R}} f(x)g(x)\,dx,$$

where $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ is your favorite field. To do this, one should observe (this is sometimes called the fundamental lemma of the calculus of variations) that if $\varphi_f(g) = 0$ holds for all g and a (continuous/locally integrable) function f, then $f \equiv 0$ (a.e.).

As the calculation above shows, we have $\varphi_{f'}(g) = -\varphi_f(g')$ for all g.

We note that the right hand side (that is, the map $g \mapsto \Phi(g')$ makes sense as soon as Φ is a linear functional on C_c^∞ (for example $\Phi = \varphi_f$), so that we **could define** the "generalized derivative" of such a functional Φ by **defining**

$$\Phi' : C_c^\infty(\mathbb{R}) \to \mathbb{K}, g \mapsto -\Phi(g')$$

Observe that this is well-defined because of $g' \in C_c^\infty$ if $g \in C_c^\infty$.

The new idea now is to define a new class of **generalized functions** as the set of all (reasonable) linear functionals on C_c^∞. As seen above, we will then be able to define a generalized derivative of these generalized functions such that it **generalizes** the ordinary derivative (i.e. $\varphi_f' = \varphi_{f'}$ if $f \in C^1$).

But all we **really** wanted to do is to be able to define arbitrary derivatives of continuous/locally integrable functions. So it would really suffice to consider the class of all (generalized) derivatives (or arbitrary order) of generalized functions from the set $\{\varphi_f \mid f \in C^0(\mathbb{R})\}$ (or replace C^0 by L^1_{loc}). Instead of these "nice" objects, the class of **all** linear functionals on C_c^∞ (in contrast to the continuous ones) would contain many pathological functionals which we want to avoid.

Instead of introducing this definition, we observe that each of the functionals φ_f is **continuous** in the sense described in your question (as you observed yourself). We also observe that this continuity property is inherited by the generalized derivative, i.e. if Φ is continuous in the sense described above, then Φ' is also a continuous linear functional in this sense (**why**?).

One can even show (this is done in Rudin's "Functional Analysis") that this definition yields the (more or less) **smallest** class of "generalized" functions such that it contains all continuous functions and arbitrary derivatives of those. More precisely, for each compact set $K \subset \Omega$ and each distribution Φ, there is some $N \in \mathbb{N}$ and continuous functions f_α such that

$$\Phi(f) = \sum_{|\alpha| \leq N} \int_\Omega f_\alpha \partial^\alpha f(x)\, dx$$

holds for all $g \in C_c^\infty(\Omega)$ with support in K.

This observation justifies the choice of functionals made above.

Also note that there is a general principle at work here: We want to have a "big" space of generalized functions. To achieve this, we choose a very "small" space ($C_c^\infty(\Omega)$) with a very strong topology/notion of convergence, i.e.

$$f_n \to f \text{ in the sense of } C_c^\infty(\Omega)$$

if and only if there is a compact set $K \subset \Omega$ such that $\operatorname{supp}(f_n) \subset K$ for all n and $\partial^\alpha f_n \to \partial^\alpha f$ uniformly for all α and define our "large" space to be the **dual space** of the "small" space, i.e. the set of all continuous linear functionals.

Share Cite Follow

answered Oct 7, 2014 at 21:15



PhoemueX

36.1k ● 3 ■ 51 ▲ 93

Would it be equivalent to say that $f_n \to 0$ in $\mathscr{D}$ if and only if $\partial^\alpha f_n \to 0$ in $\mathscr{D}$? My professor asserted the topology as such, and used it to prove the continuity of distributions, by noting that if $\{\psi_n\} \subset \mathscr{D}$ goes to 0, then the distribution is continuous if and only if it also goes to 0. – Anthony Peter Oct 8, 2014 at 1:03

Yes, the definition easily shows that $f_n \to 0$ in $\mathscr{D}$ iff $\partial^\alpha f_n \to 0$ in $\mathscr{D}$ **for all** α. Here "$\Leftarrow$" follows using $\alpha = 0$. One can also show (using only the definitions) that a functional Φ on C_c^∞ is a distribution (i.e. coninuous) if and only if $\Phi(f_n) \to 0$ as soon as $f_n \to 0$ in $\mathscr{D}$.

Note that $f_n \to 0$ in $\mathscr{D}$ is equivalent to $\operatorname{supp}(f_n) \subset K$ for some fixed compact $K \subset \Omega$ together with $\partial^\alpha f_n \to 0$ **uniformly** for all α. – PhoemueX Oct 8, 2014 at 13:31

Add a comment

u must log in to answer this question.

Start asking to get answers

Find the answer to your question by asking.

Ask question

Explore related questions

real-analysis analysis distribution-theory

See similar questions with these tags.

e Inc; user contributions licensed under CC BY-SA . rev 2025.4.17.25340